

Analysis of automobile (Motorbike) sales trends for revenue maximization and inventory optimization.

A Final Submission Report for the BDM Capstone Project

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Executive Summary

This proposal outlines a plan to address the business challenges of the TVS Bike showroom in Purulia District, West Bengal, India. The project aims to understand their current problems and facilitate growth. Data collection began on January 2nd, gathering information from stakeholders and compiling relevant data from January 2nd to January 7th. This involved organizing Excel files containing bike model data and sales data by year. Additionally, profit details were requested to address incomplete purchase data. Data was then cleaned and transformed using Python, ensuring accuracy. Power BI will be used for comprehensive data analysis and visualization, focusing on SKU performance, customer behavior, pricing, costs, and tax implications. Predictive models will be developed to forecast demand and preferences. The ultimate goal is to provide actionable recommendations through intuitive dashboards to improve customer targeting and overall business performance.

Detailed Explanation of Analysis Process/Method :

Data Collection and Preprocessing:

- Utilized Power BI for Data Analysis and Visualization.
- Employed Python with the scikit-learn library for preprocessing.
- Conducted preprocessing tasks within VS Code as a project.
- Handled missing data by utilizing pandas DataFrame.
- Read yearly sheet data of the organization using pandas DataFrame.
- Applied scaling to all columns.
- Concatenated all individual DataFrames into one master DataFrame.

Data Modification:

- Modified certain columns as required.
- Checked for missing values in the master DataFrame.
- Filled missing values with appropriate data.
- Handled datetime columns for consistency.

Profit Calculation:

- Since there was no profit column, consulted Mr. Rajeev about product group-wise profit.
- Performed back calculation using selling and profit values.
- Added a new column for purchased value based on calculations.

Export and Import:

- Saved the final DataFrame into CSV format for further analysis.

Visualization in Power BI:

- Imported the master sales data into Power BI for visualization.
- Created approximately six dashboards covering various aspects such as overview, sales, product, customer, tax and prices, and top performance.

Dashboard Creation:

- Overview Dashboard: Provides an overall trend of the data over time.
- Sales Dashboard: Visualizes sales performance metrics.
- Product Dashboard: Analyzes product-related data.
- Customer Dashboard: Focuses on customer-related insights.
- Tax and Prices Dashboard: Explores tax and price trends.
- Top Performance Dashboard: Highlights top-performing aspects of the data.

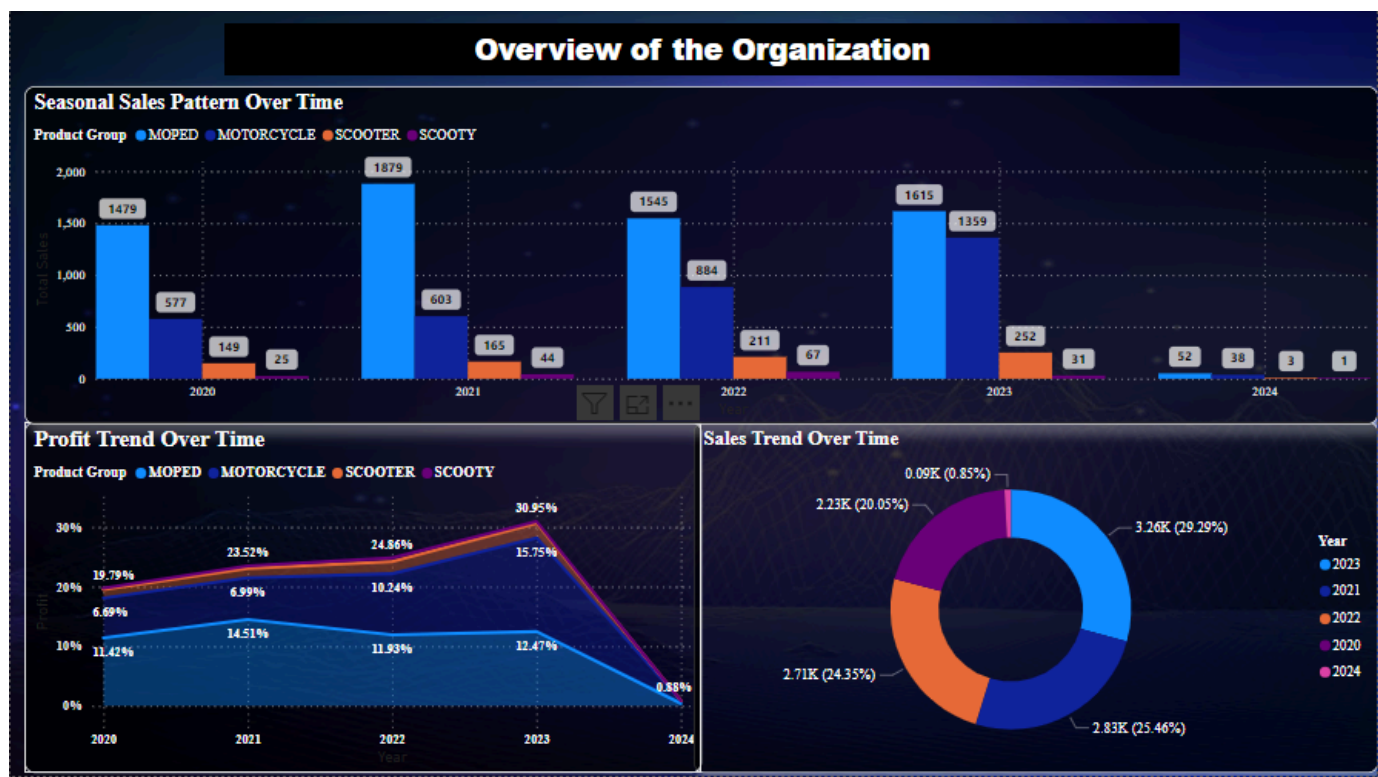
Creating separate dashboards for visualizing the entire dataset provides a versatile and interactive approach for decision-making and future forecasting. These interactive dashboards allow for the selection of specific years to view individual year-wise data, as well as an option to view overall data by selecting 'all' years. This interactivity enhances the usability of the dashboards, facilitating easy exploration and analysis of the data. Moreover, this method offers the advantage of real-time updates. By linking the dashboards to the dataset, any updates or modifications to the data will automatically reflect in the visualizations. This ensures that decision-makers are always working with the most current information, without the need for manual intervention. Additionally, while preprocessing was conducted using Python libraries, employing interactive dashboards for visualization proves more suitable. It enables potential automation of the process for future raw data, streamlining the analysis pipeline. This adaptability and efficiency make the approach highly practical for ongoing data

analysis needs. Ultimately, through these interactive dashboards, users can delve into descriptive statistics and uncover key findings, empowering informed decision-making and strategic planning.

Results and Findings (Visual Representation) :

Important Note__ All Power BI dashboards are dynamic and presentable. The examples provided are snapshots of each dashboard, illustrating a single year and a single product. The overall statistics are based on these dynamic dashboards.

Overview Dashboard :



From the above dashboard I can say below statistics:

MOPED

- Moped sales showed resilience in 2020 and 2021 despite the impact of COVID-19, but experienced a notable decline of 17% in 2022. Sales rebounded strongly in 2023 with a year-to-date growth of 10%. Profitability consistently grew over the four-year period, maintaining double-digit increases from 11% in 2020 to 13% in 2023.

- **Forecasting:** The moped segment is expected to continue growing steadily, potentially achieving a 13% profit margin. To sustain this growth, the strategy includes ongoing innovation, expanding market presence, and maintaining high product quality and customer satisfaction.

MOTORCYCLE

- Motorcycle sales steadily increased from 577 in 2020 to 1359 in 2023, accompanied by a significant rise in profit margins from 7% to 16%. The current status reflects substantial improvement and strong market penetration.
- **Forecasting:** Future projections suggest a potential 17-18% profit margin. To capitalize on this growth, the strategy emphasizes continuous development of new models with enhanced features, exploring new markets, and increasing market share. Additionally, implementing customer loyalty programs and enhancing after-sales services are recommended.

SCOOTER

- Scooter sales grew incrementally from 149 in 2020 to 252 in 2023, with a moderate increase in profit margin from 2% to 3%. The current status indicates gradual improvement but remains lower compared to other categories.
- **Forecasting:** The forecast anticipates achieving a 3-4% profit margin. To achieve this, strategies include redesigning products with improved features, enhancing visibility through promotions and discounts, and implementing cost-saving measures to improve profitability.

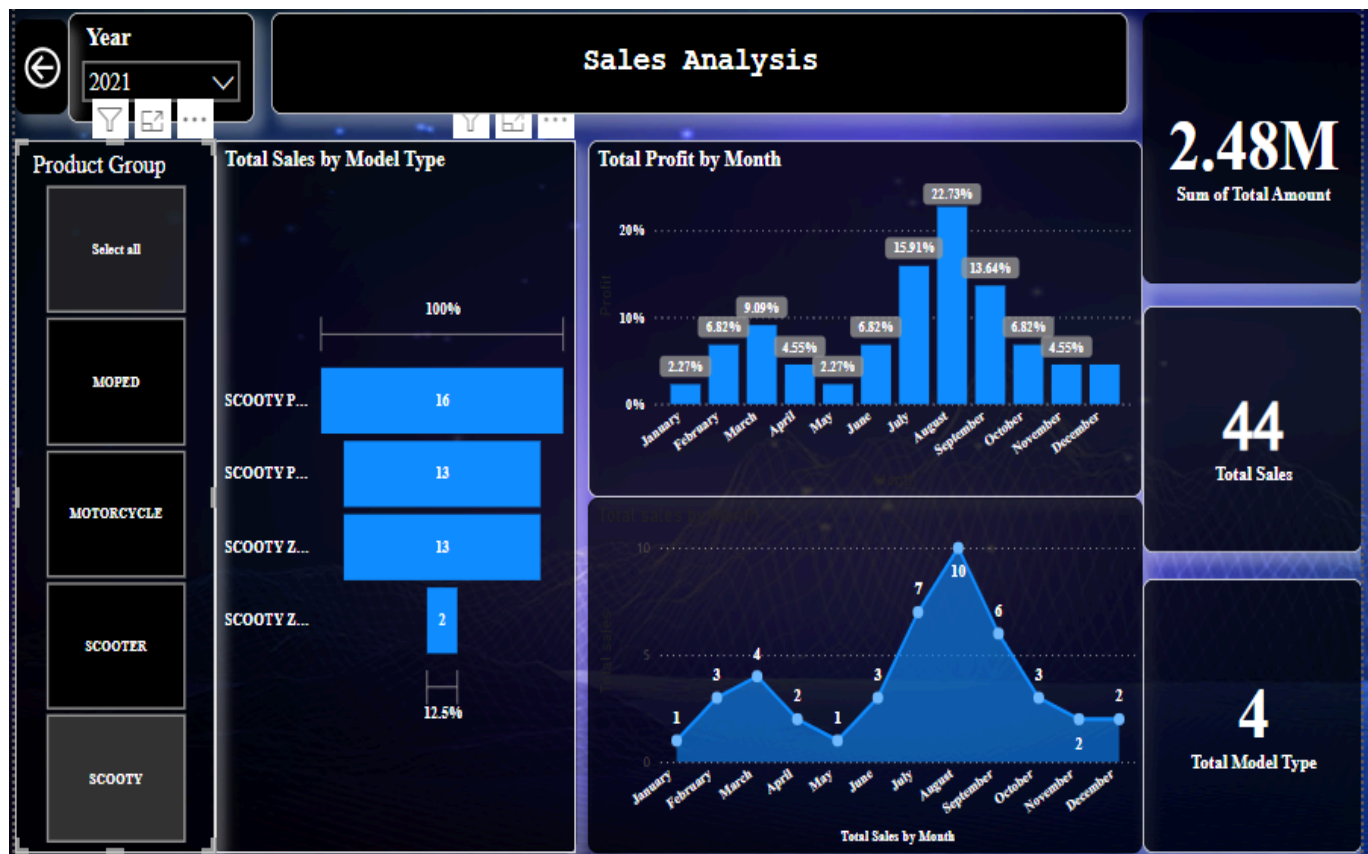
SCOOTY

- Scooty sales fluctuated from 25 in 2020 to 67 in 2022, with a slight decline to 38 in 2023. The current status shows a marginal increase in profitability.
- **Forecasting:** Changes include conducting a comprehensive market analysis, considering product repositioning or targeting different market segments, and enhancing product features based on customer feedback.

Sales Trend :

- ➔ In 2020, our analysis began with capturing 20% of the overall sales volume. This foundational year was marked by establishing key strategies and understanding market dynamics. The focus was on building a strong customer base and enhancing product visibility. By 2021, the sales volume increased to 25%, showing a 5% growth despite the challenges posed by the COVID-19 pandemic. The increase can be attributed to adaptive strategies such as leveraging online sales channels, enhancing digital marketing efforts, and introducing flexible purchase options.
- ➔ In 2022, the sales volume remained steady at 25%, maintaining the growth achieved in the previous year. This stability indicates that our strategies continued to resonate well with customers. Focused efforts on customer retention, product quality, and after-sales services helped in sustaining this performance. Additionally, consistent marketing campaigns and promotions played a role in keeping the sales volume steady.
- ➔ By 2023, the sales volume saw a significant increase, reaching 30%. This 5% growth reflects the culmination of successful initiatives and a robust market demand. Continued investment in product innovation, enhanced customer satisfaction, and effective marketing strategies contributed to this increase. The introduction of new models and features, coupled with targeted promotions, helped in capturing a larger market share.

Sales Analysis Dashboard :



From the above dashboard I can say below statistics:

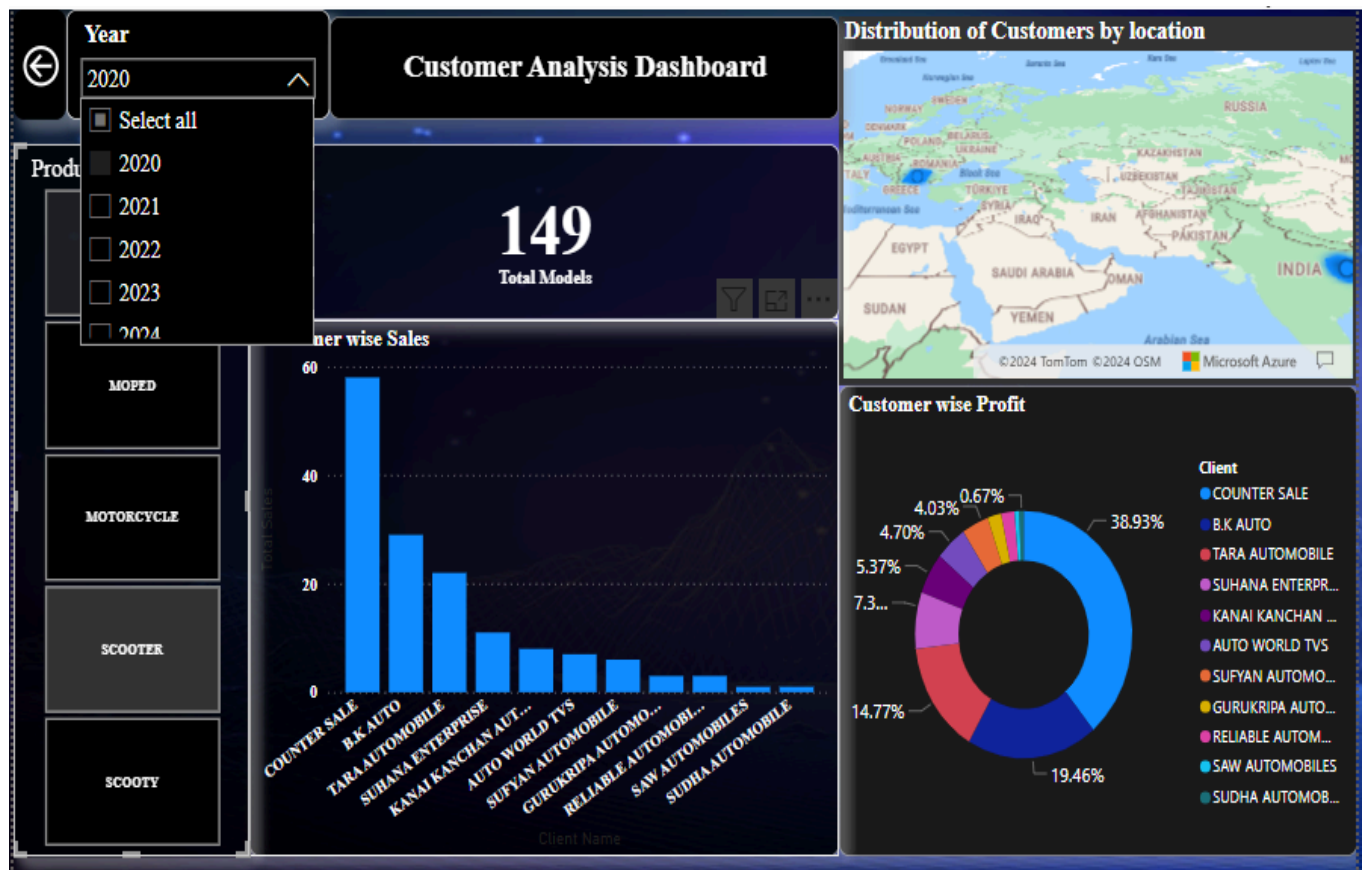
- October is consistently the peak month for sales and profits for all products, with profit margins ranging from 12-14%. This indicates a strong seasonal demand likely driven by factors such as festivals, holidays, or end-of-year promotions. The festive season often sees a surge in consumer spending, as people look to make significant purchases during this period. Additionally, end-of-year bonuses and the general celebratory mood contribute to higher sales. Capitalizing on this, it's crucial to ensure that inventory levels are sufficient to meet the increased demand, and marketing efforts are intensified to capture the attention of potential buyers.
- May is the lowest month for sales and profits for Moped and Motorcycle. This suggests a potential seasonal dip in demand, possibly due to factors like summer vacations or climatic conditions. During this period, the heat might deter people from buying two-wheelers, or they might be allocating their budget to other expenditures

like travel. To counteract this, offering special summer promotions or bundling deals can help stimulate interest and boost sales. Engaging customers with limited-time offers or finance schemes can also be effective strategies to mitigate the seasonal slump.

- ➔ February is the lowest month for Scooter sales, and June is the lowest for Scooty, indicating different seasonal lows for these products. This variation could be due to differing consumer behavior or market conditions. For instance, in February, post-holiday financial constraints may lead to reduced discretionary spending. Conversely, in June, the onset of the monsoon season might impact the sale of Scooties. Tailored marketing campaigns focusing on the unique selling points of these vehicles, coupled with attractive discounts, can help maintain a steady sales flow. Collaborating with financial institutions to provide easy loan options during these months might also attract budget-conscious consumers.
- ➔ To maximize sales and profit margins, ramping up production, inventory, and marketing efforts in the months leading up to October is essential. Leveraging the predictable peak demand, businesses can plan extensive advertising campaigns, both online and offline, to create buzz around their products. Special festive offers, early bird discounts, and value-added services can enhance customer appeal. Additionally, strengthening the supply chain to ensure timely availability of products will help meet the heightened demand without any stockouts.
- ➔ For the low-demand months, targeted promotions or discounts might help in maintaining steady sales and clearing out inventory. Creating a sense of urgency with time-limited offers can encourage quicker purchases. Engaging with customers through personalized marketing, loyalty programs, and referral bonuses can also drive sales during these periods. Furthermore, offering flexible financing options can ease the purchasing decision for potential buyers.
- ➔ As a TVS dealer, exploring the introduction of new models or features before October can attract more customers and create excitement in the market. Organizing launch events, offering demo rides, and providing exclusive previews can generate significant interest and anticipation among consumers. Highlighting the innovative aspects of the new TVS models through comprehensive marketing strategies, including social media campaigns, influencer partnerships, and customer testimonials,

can further enhance the product's appeal. Additionally, ensuring that the new models are competitively priced and offer better value for money can position them as attractive options for potential buyers. This strategic approach can help leverage the peak sales season, driving higher customer engagement and boosting overall sales performance.

Customer Analysis Dashboard :



From the above dashboard I can say below statistics:

- **Counter sales (individual clients)** dominate with the highest sales, making them a crucial focus for our business strategy. Counter sales contribute significantly to

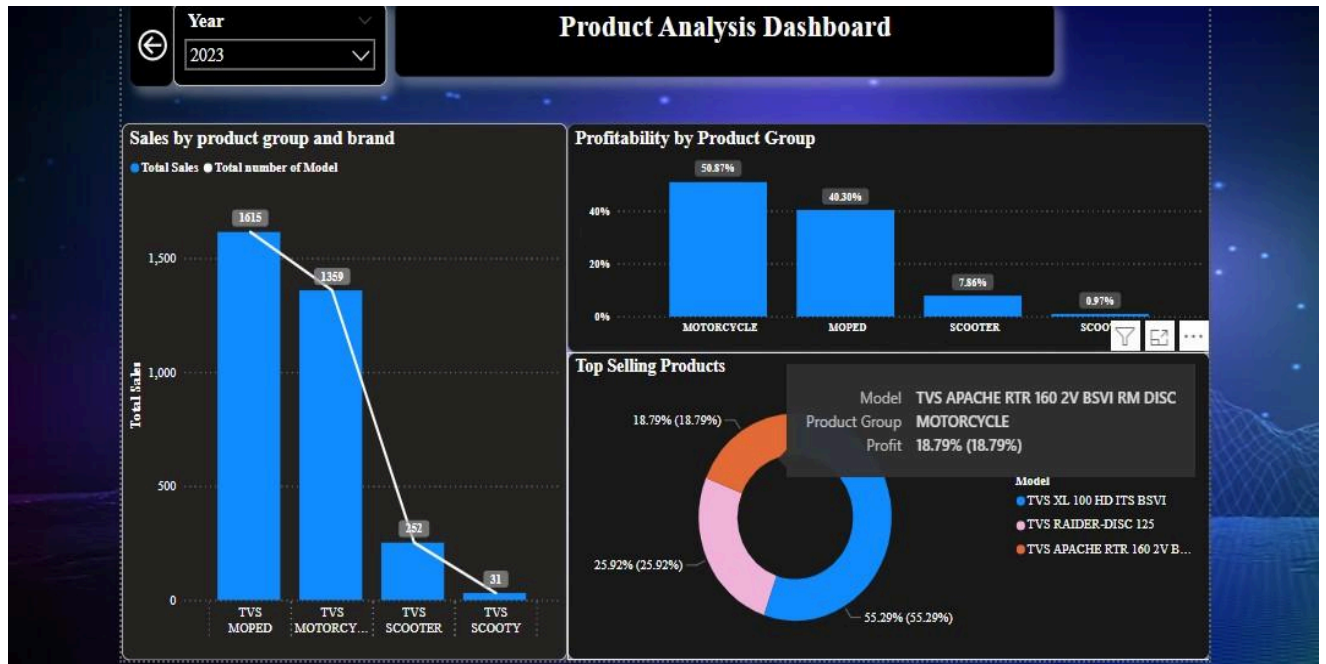
overall profits, accounting for 38% of the total profit for Mopeds, 53% for Motorcycles and Scooters, and approximately 58% for Scooties. This substantial contribution underscores the importance of maintaining strong relationships with individual clients and continuously enhancing the customer experience at the counter. By providing exceptional service, personalized recommendations, and attractive deals, we can further strengthen this revenue stream.

- ➔ **Sub-dealers** also play a vital role in our sales ecosystem. **B.K. Auto, Sufyan Automobile**, and **Suhana Enterprises** are key sub-dealers, particularly notable for their substantial Moped sales. Among them, only B.K. Auto shows substantial Motorcycle sales as a sub-dealer, highlighting its significance in this segment. Additionally, Tara Automobile and B.K. Auto exhibited considerable Scooter and Scooty sales, indicating their importance in these categories. Maintaining robust partnerships with these sub-dealers is essential to ensure consistent sales performance. Offering tailored support, exclusive deals, and incentives can help in nurturing these relationships and driving mutual growth.
- ➔ The remaining clients, although contributing smaller shares to the overall profit, still play a role in our business landscape. Each of these clients contributes up to 8% of the total profit, which, when aggregated, forms a significant portion of our revenue. By implementing dynamic pricing strategies, we can maintain profit margins and encourage sales during periods of low demand. This approach involves adjusting prices based on market conditions, competitor pricing, and demand fluctuations, ensuring that we remain competitive and profitable throughout the year.
- ➔ It is crucial to pay close attention to major clients like Counter Sale, B.K. Auto, Tara Automobile, and Sufyan Automobile, as they significantly impact both sales and profits. Regularly engaging with these key clients to understand their needs and feedback can lead to improved service and stronger business relationships. Developing targeted marketing campaigns, loyalty programs, and customized offers for these major clients can further enhance their satisfaction and loyalty, ultimately driving higher sales and profitability.

Geographic Distribution(Optional Here for future)

The map shows a wide distribution of customers, primarily across the Middle East and South Asia. This indicates a broad market presence in these regions.

Product Analysis Dashboard :



From the above dashboard I can say below statistics:

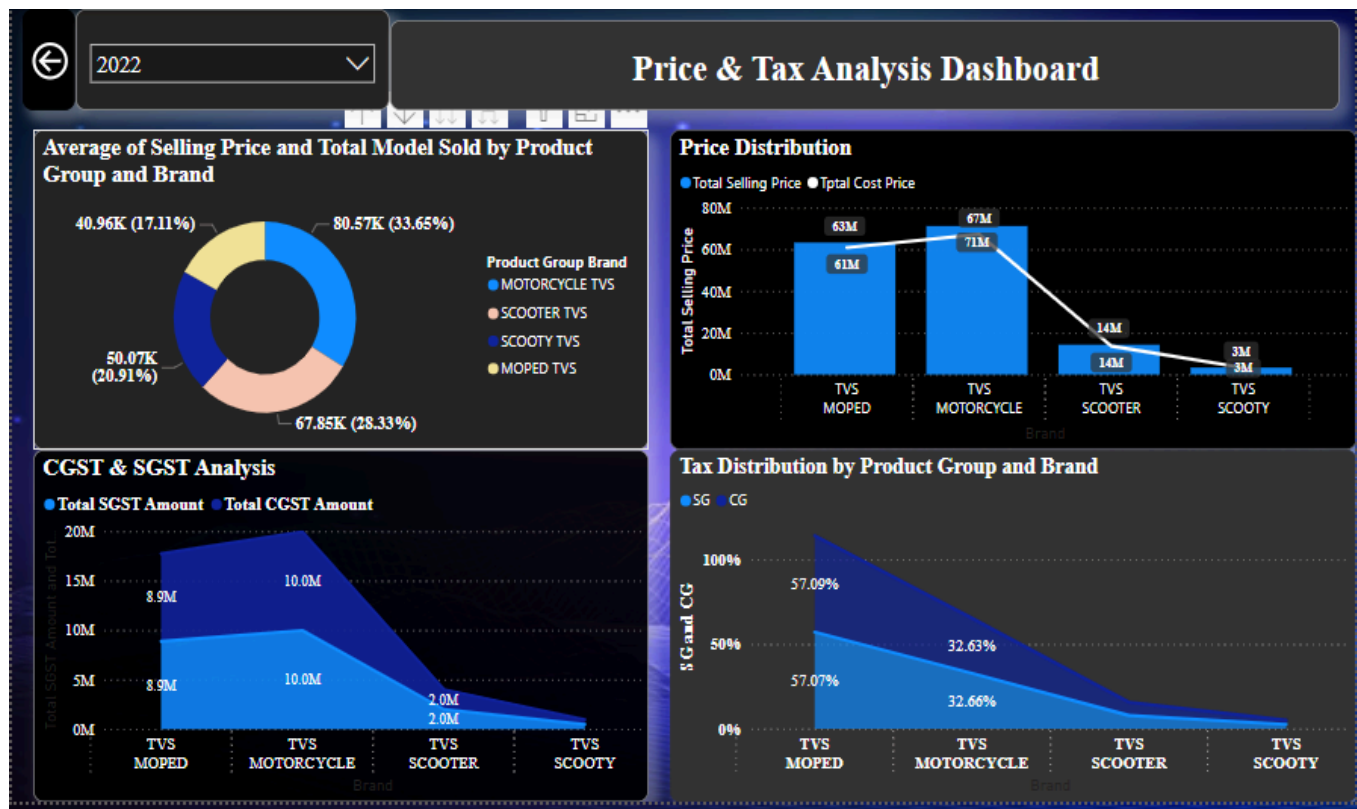
Product and Profit wise statistics

- **TVS Moped:** Leading in total sales with approx 7K units sold, indicating high market demand and popularity in this category in total from 2020 to 2023. Highest profitability at near about 51%, making it the most profitable product group.
- **TVS Motorcycle:** Second in total sales with approx 4K units sold, showing a strong presence in the market. Second highest in profitability at 41%, indicating a significant contribution to overall profits.
- **TVS Scooter:** Indicating moderate demand. Lower profitability at 8%, suggesting a need for strategy reevaluation.
- **TVS Scooty:** The lowest in sales suggesting either low demand or limited availability. Lowest profitability at highlighting it as a weak performer in terms of profit.

Product Model wise statistics

- ➔ TVS XL 100 HD ITS BSVI Captured 70% of the market in 2020, 72% in 2021, 65% in 2022, and 56% in 2023. This model remains a significant player. However, to sustain its position, TVS may need to innovate and enhance this model, possibly introducing new features or variants to attract more customers.
- ➔ TVS RAIDER-DISC 125 held 26% of the market in 2020, 23% in 2022, and 26% in 2023. With an increasing market share over recent years, this model shows strong growth potential. Continued focus on marketing, maintaining product quality, and possibly expanding the range could help capitalize on its rising popularity.
- ➔ TVS APACHE RTR 160 2V BSVI RM DISC Captured 19% of the market in 2020, 16% in 2021, and 19% in 2023. The relatively stable market share indicates a loyal customer base. By enhancing its features or introducing new variants, TVS can potentially increase its market share.
- ➔ TVS XL 100 COM ITS-BSVI held 14% of the market in 2021 and 13% in 2022. has seen a decline in market share and was not among the top sellers in 2023. This model may need a reevaluation of its market strategy or significant improvements to regain its market position.

Price & Tax Analysis Dashboard :

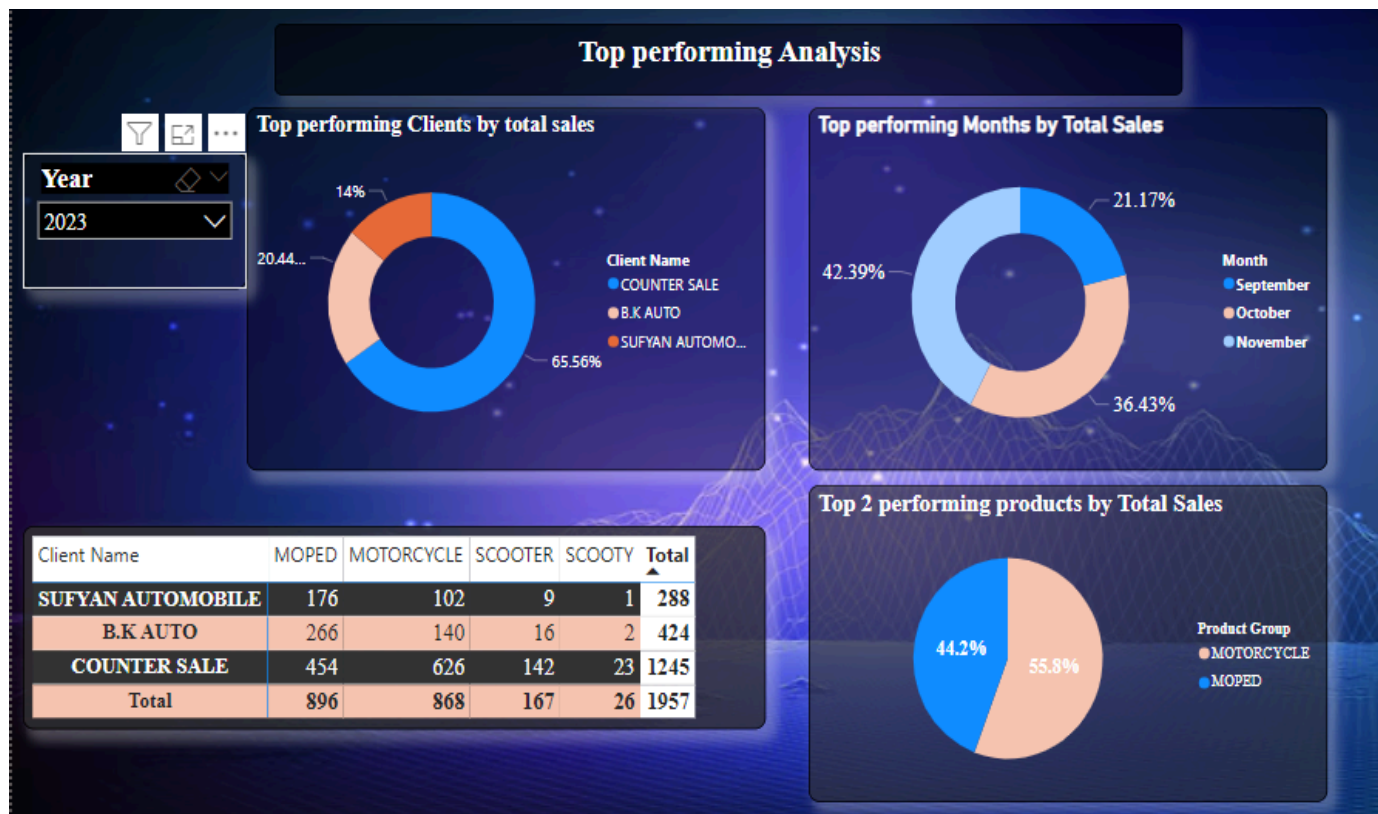


From the above dashboard I can say below statistics:

- Motorcycles (TVS) dominate the market, accounting for approximately 40% of the total sales, with an average selling price of 102K; the total SGST and CGST amounts are highest for Motorcycles (TVS) at approximately 50M for both taxes, with SG and CG contributions at approximately 32% for this product group.
- Scooters (TVS) hold around 26% of the total sales, with an average selling price of 65K; the total SGST and CGST amounts for Scooters (TVS) reflect their market share, with proportionate SG and CG contributions.
- Mopeds (TVS) account for about 16% of the total sales, with an average selling price of 40K; the total SGST and CGST amount for Mopeds (TVS) is 36M, with SG and CG contributions approximately at 60%.
- Scooty (TVS) has the lowest market share at around 20% of the total sales, with an average selling price of 49K; the total SGST and CGST amounts for Scooty (TVS)

are the lowest, reflecting their sales volume, with proportionate SG and CG contributions.

Top Performer Analysis Dashboard :

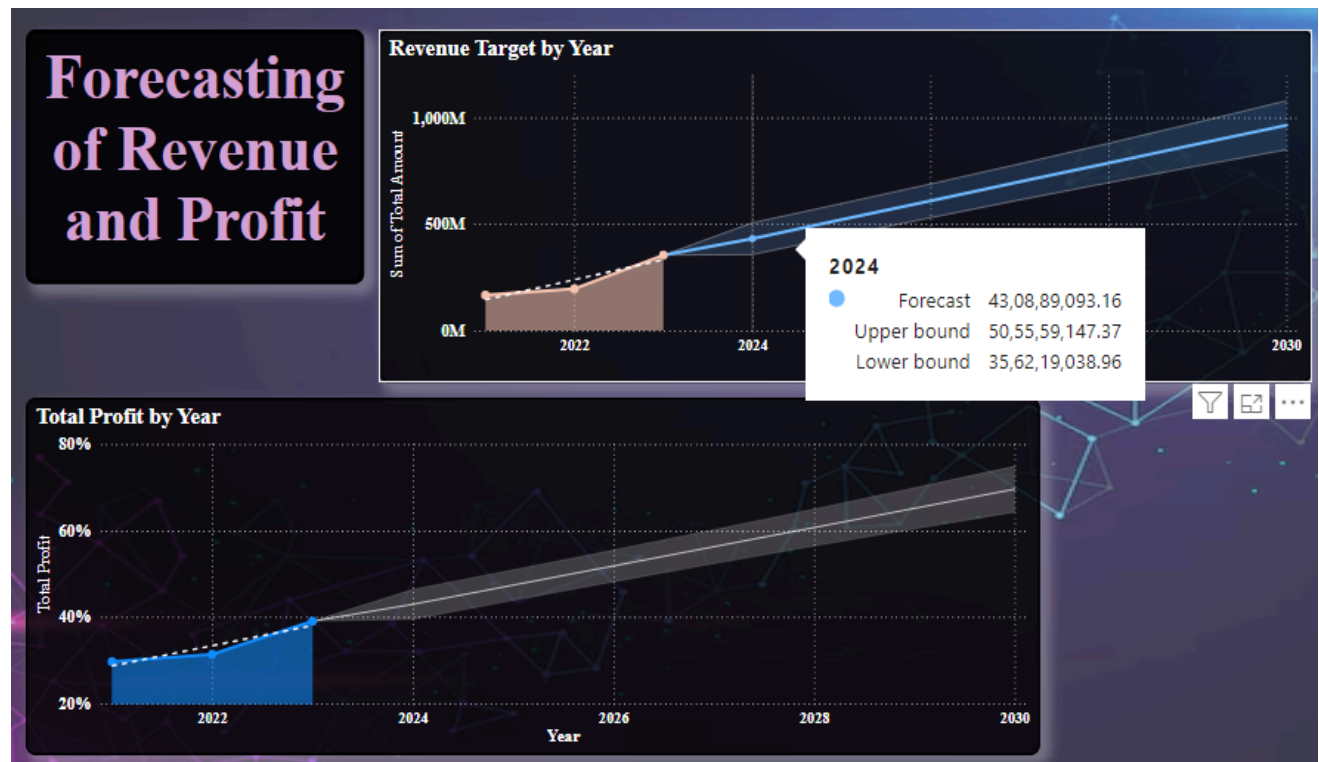


From the above dashboard I can say below statistics:

- Top performing clients are Counter sale(Individual clients)
- Top performing clients are B.K. Auto and Sufyan Automobile who are also sub dealers
- In the months of September, October and November are top performing months throughout the past years .
- Top products are Moped and Motorcycle throughout the past years
- Focus marketing efforts on **Counter Sale** to leverage their dominant market position.

- Implement targeted promotional campaigns for **B.K Auto** to increase market share, especially for Motorcycles.
- Capitalize on peak sales months (November and October) with special promotions and discounts to maximize sales.

Forecasting Analysis Dashboard :



From the above dashboard I can say below statistics:

- The provided forecasting data for the TVS showroom in Purulia indicates that the revenue target for 2024 is ₹50M, with a forecasted revenue of ₹43M and a lower bound of ₹35M. This shows a strong potential for achieving the showroom manager's ambitious sales target of ₹50M, aligning closely with the upper bound of the forecast.
- The profit margin forecast also shows promising growth, with total profit expected to increase significantly from its current state in 2022. The trend indicates a positive trajectory, suggesting that with the right strategies, the showroom can not only achieve but potentially exceed its revenue and profit targets by 2030.

Interpretation of Results and Recommendation :

- Moped segment expected to achieve a 13% profit margin through innovation, market expansion, and high product quality.
- Future projections suggest a 17-18% profit margin with a focus on new model development, market exploration, customer loyalty programs, and after-sales service enhancement.
- Achieving a 3-4% profit margin by redesigning products, promoting visibility, and implementing cost-saving measures.
- Conduct market analysis, consider product repositioning, and enhance features based on customer feedback.
- Ramp up production, inventory, and marketing before October to maximize sales and profits.
- Use targeted promotions during low-demand months to maintain steady sales and clear inventory.
- Introduce new models or features before October to attract customers.
- Implement dynamic pricing during low-demand periods to maintain profit margins and encourage sales.
- Focus on major clients like Counter Sale, B.K. Auto, and Tara Automobile as they significantly impact sales and profits.
- Innovate and enhance TVS XL 100 HD ITS BSVI, capitalize on the strong growth potential of TVS RAIDER-DISC 125 with continued marketing and quality maintenance, enhance features or introduce new variants for TVS APACHE RTR 160 2V BSVI RM DISC, and reevaluate the market strategy or improve TVS XL 100 COM ITS-BSVI to sustain or increase market positions.
- Focus marketing efforts on TVS motorcycles to leverage market dominance.
- Introduce promotional campaigns for TVS Scooty to boost sales and market share.
- Monitor cost price trends and adjust selling prices to maintain profitability.
- Keep an eye on tax regulations for optimal pricing and tax compliance.
- Focus marketing on Counter Sale to leverage their dominant market position.
- Implement targeted campaigns for B.K Auto to increase motorcycle market share.
- Capitalize on peak sales months with special promotions and discounts.

Summary of Project :

The proposed project aims to address key challenges faced by the TVS showroom in Purulia by leveraging sales data and external insights to optimize SKU availability, develop targeted pricing strategies, and maximize profitability. By analyzing sales trends and external factors, the project will enable proactive inventory management, ensuring optimal stock levels and readiness for market changes. Targeted customer segmentation and personalized pricing models will enhance customer satisfaction and revenue. Dynamic pricing strategies and cost reduction measures will further improve profitability. Focused marketing efforts and product innovations will drive market expansion and customer loyalty, positioning the showroom for sustained growth and achievement of the ₹500,000,000 sales target. This comprehensive approach promises not only to stabilize sales fluctuations but also to fulfill customer aspirations and drive long-term business success in Purulia.

Limitations

Time-factor impact not consider for this analysis
